Day 1 — Modern LLM Foundations

> 🎯 **Goal**: Understand how LLMs work from tokens to training — the foundation for everything in this course.

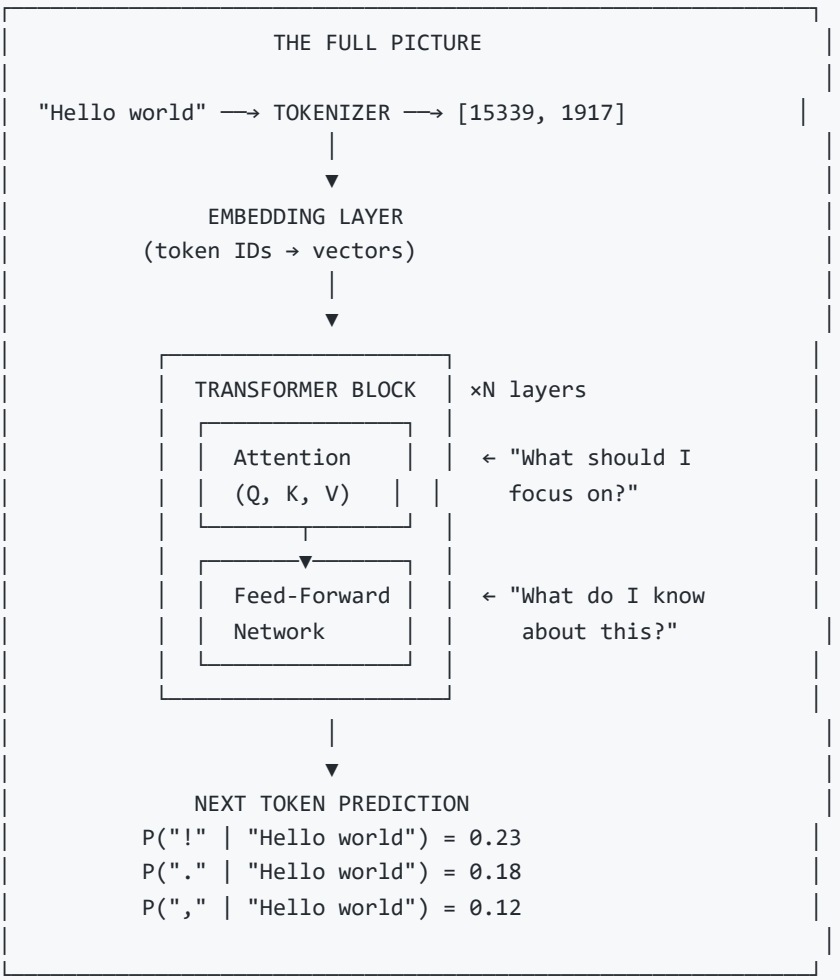
> ⏱️ **Time**: 3.5–4.5h total (across 3 parts)

> 📦 **Deliverable**: Tokenization explorer + sampling visualizer + ability to explain the full Transformer pipeline

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Overview

Before you can build agents, you need to understand the engine they run on. This day covers the complete LLM pipeline — from raw text to trained model to generated output.



Parts

This day is split into three focused parts:

Part	Topic	Time	What You'll Learn
Part A	Tokens & Embeddings	~1.5h	What tokens are, how BPE works, why embeddings capture meaning
Part B	Transformer & Attention	~2.5h	Self-attention (Q/K/V), multi-head attention, FFN, residual connections
Part C	Training & Inference	~2h	Positional encoding, pre-training/SFT/RLHF pipeline, sampling strategies
Appendix	Advanced Techniques	Optional	MoE, GQA, Flash Attention, LoRA, quantization, scaling laws

Recommended Order

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Part A (foundation) → Part B (architecture) → Part C (training & generation)
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Each part is self-contained with its own exercises, but concepts build on each other.

Key Questions You'll Be Able to Answer

After completing Day 1, you should be able to answer:

1. **Tokenization**: Why isn't a token always a word? What is BPE?
2. **Embeddings**: How do computers represent meaning as numbers?
3. **Attention**: What are Q, K, V and why do we need all three?
4. **Architecture**: What does each component of a Transformer block do?
5. **Training**: What's the difference between pre-training, SFT, and RLHF?
6. **Inference**: How does temperature affect generation? What is KV-cache?

▶ 📝 **Answer Key (click to expand)**

✓ Day 1 Deliverable Checklist

- [ ] Explored tokenization with tiktoken
- [ ] Can draw the Transformer architecture from memory
- [ ] Built a sampling strategy visualizer
- [ ] Can explain: tokens, embeddings, Q/K/V, attention, FFN
- [ ] Can explain: pre-training vs SFT vs RLHF
- [ ] Can explain: temperature, top-k, top-p, KV-cache

Start → Part A: Tokens & Embeddings

Next Day → Day 2: Advanced Prompt Engineering